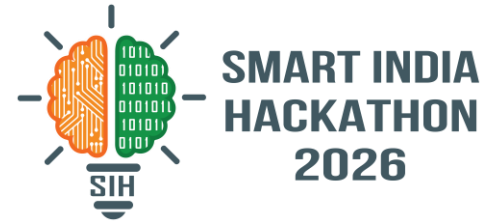
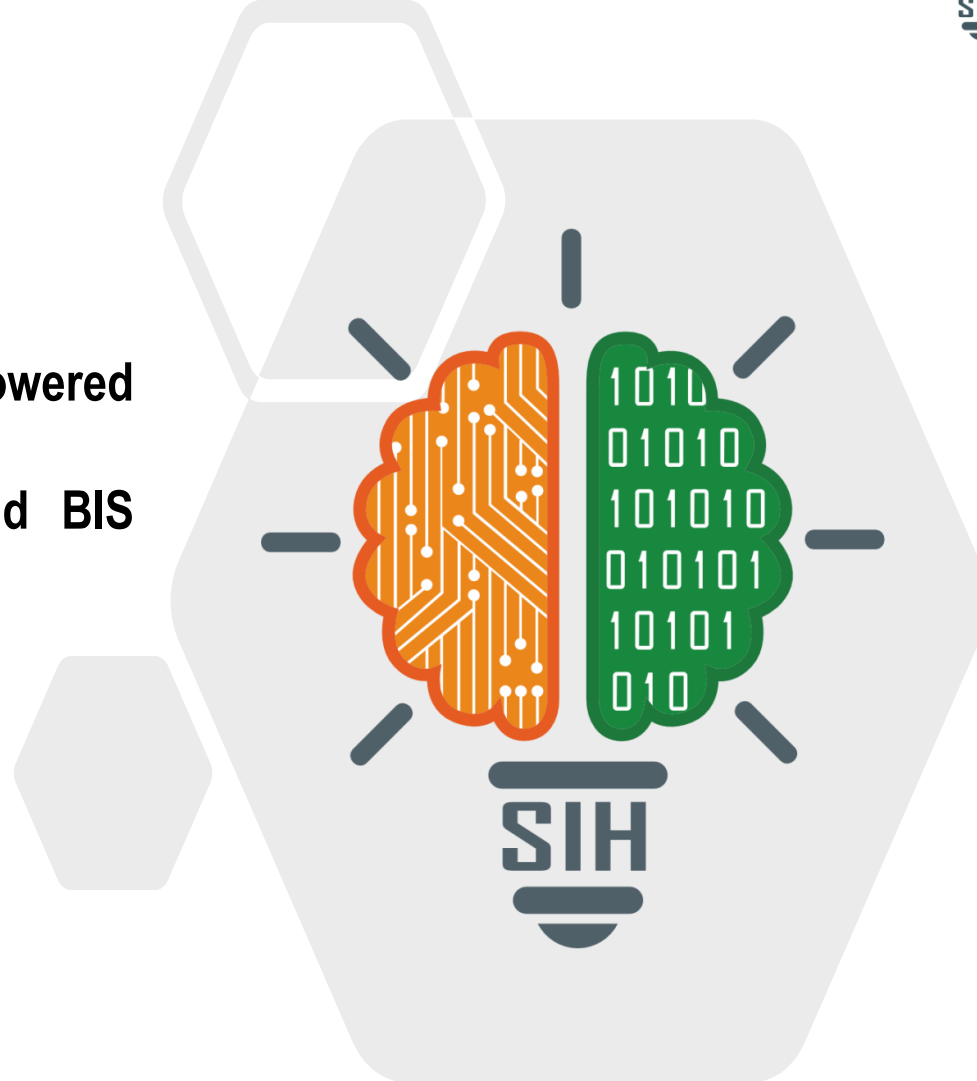


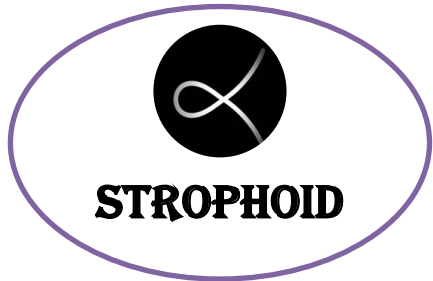


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26107
- **Problem Statement Title-** AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name-** STROPHOID





BIS AI-SAATHI



SMART INDIA HACKATHON 2026

“Your AI Guide to Indian Standards & BIS Certification”

PROBLEM

In India, the Bureau of Indian Standards (BIS) has enforced **350+ mandatory Quality Control Orders (QCOs)**, which include cement, steel, electronics, helmets, pressure cookers, and toys.

For Industry /MSMEs: Reading complex legal standards, understanding factory inspection rules, and following testing protocols is so difficult that getting certification takes months.

For Common Consumers: Millions of fake ISI marks and substandard products are being sold in the market. The common citizen does not have any tool to take a photo and find out in 2 seconds whether the mark is real or fake.

SOLUTION

BIS Saathi is a **Multimodal Agentic AI Assistant** that solves both problems on a single portal. It is not just a simple chatbot, but a complete **compliance operating system** that is grounded in official Gazette notifications and Indian Standards (IS Codes), detects fake marks using computer vision, and facilitates the filing of legal grievances.

WHY DIFFERENT?

- AI maps product to exact Indian Standard — no manual search
- Natural language input — English or Hinglish
- License Checker — verify in 2 seconds

KEY VALUE

- Saves time** — seconds not hours
- Prevents penalties** — verify first
- One platform** — all stakeholders



Helmets



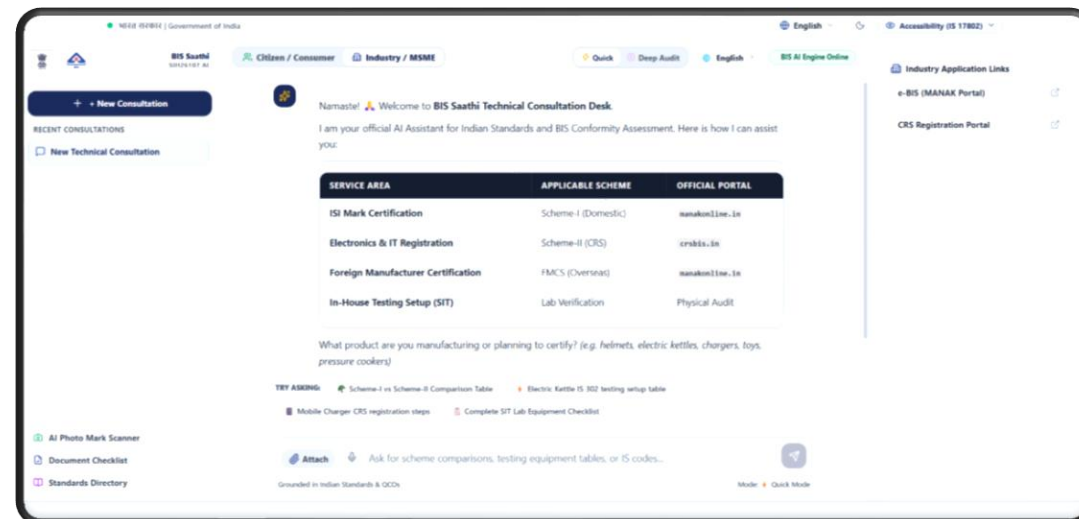
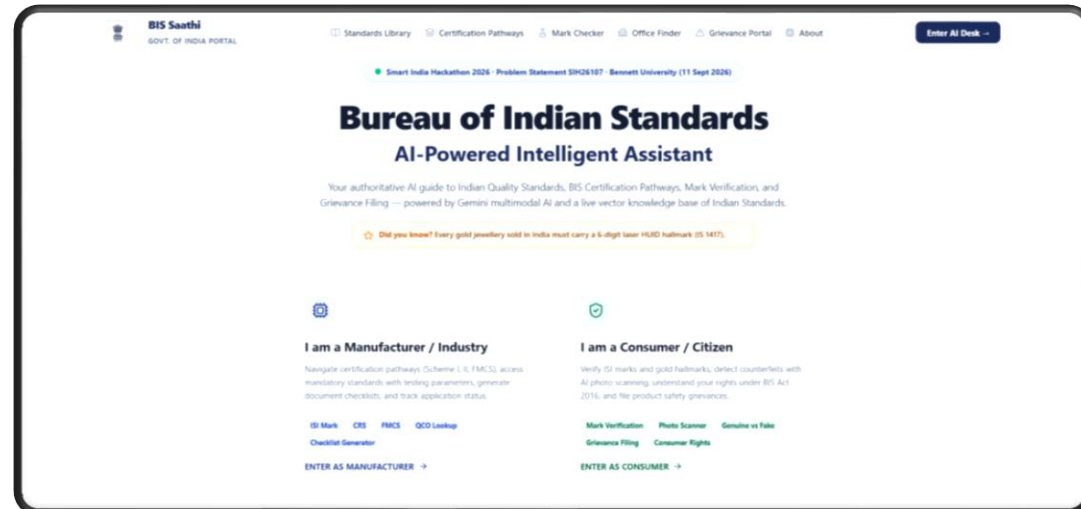
Chargers



Toys

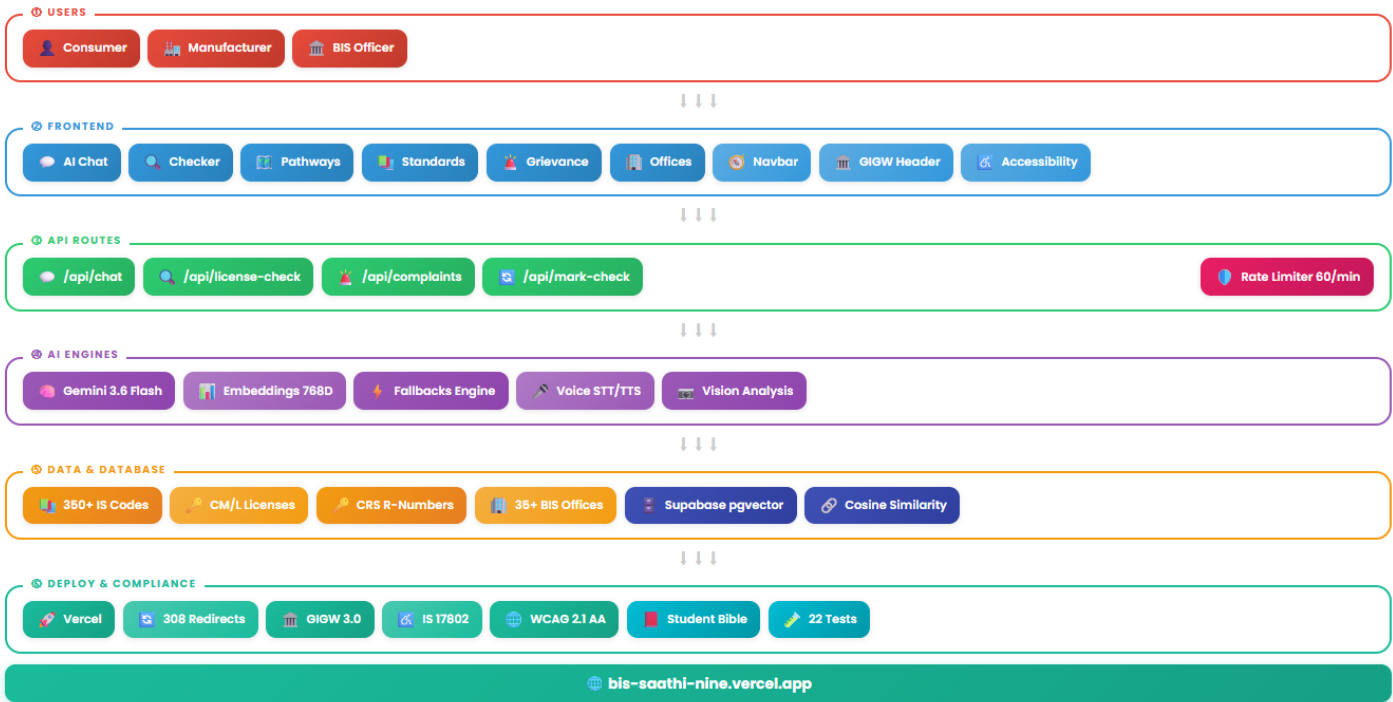


LED





----- System Architecture -----



----- Process Flowchart -----

Tech Stack Used:

Next.js 16	FRONTEND	✓
React 19	FRONTEND	✓
TypeScript	LANGUAGE	✓
Tailwind CSS 4	STYLING	✓
Google Gemini 3.6 Flash	AI ENGINE	✓
text-embedding-004	EMBEDDINGS	✓
Supabase PostgreSQL	DATABASE	✓
pgvector	VECTOR SEARCH	✓
Vercel	DEPLOY	✓
PostCSS	CONFIG	✓
ESLint	LINTING	✓
GIGW 3.0 + IS 17802	ACCESS	✓



FEASIBILITY AND VIABILITY



SMART INDIA
HACKATHON
2026

• TECHNICAL FEASIBILITY (GOVT DEPLOYMENT)

- Cloud-native architecture — deploy on NIC or MeitY cloud infrastructure
- RAG + pgvector — scales to handle 350+ standards and millions of queries
- GIGW 3.0 compliant — meets mandatory govt website guidelines
- Multi-model cascade — 99.9% uptime with automatic failover

• OPERATIONAL FEASIBILITY (DAILY USE)

- Citizens get instant BIS guidance — reduces office footfall by 60%+
- MSMEs self-serve certification queries — no manual officer intervention
- Grievance portal auto-generates docket IDs — streamlined complaint tracking
- Hindi + regional language support — pan-India accessibility

• ECONOMIC VIABILITY (COST-BENEFIT)

- MVP cost under ₹500/month — negligible vs govt budget allocation
- Reduces manual query handling — saves officer time and infrastructure cost
- Open-source stack — no vendor lock-in, no licensing fees
- Scales linearly — cost grows with usage, not upfront investment

• RISK ASSESSMENT (GOVT DEPLOYMENT)

- AI accuracy on niche standards — may give wrong IS codes
- Data sovereignty — user queries stored on third-party cloud
- API dependency — Gemini service outage affects availability
- Legal liability — AI-generated advice vs official BIS guidance

• MITIGATION STRATEGIES

- Regulatory Fallbacks Engine — offline mode, no AI dependency
- Deploy on NIC cloud — full data sovereignty within India
- Curated Indian Standards DB — verified data, not web-scraped
- Disclaimer on every response — "for reference, not legal advice"

• SCALABILITY & FUTURE SCOPE

- Integrate real BIS API — live license verification from manakonline.in
- Expand to all 350+ QCOs — full Indian Standards catalog
- WhatsApp chatbot integration — reach 500M+ Indian users
- Analytics dashboard — track compliance trends for policy decisions

GOVT ADOPTION VIABILITY



88%



IMPACT AND BENEFITS



SMART INDIA
HACKATHON
2026

----- What Happens When Government Adopts BIS AI Saathi -----

500M+

Citizens Protected by
BIS Regulations

350+

Products Under
Mandatory QCOs

60%

Less BIS Office
Footfall Expected

10x

Faster Certification
Guidance



SOCIAL IMPACT (PUBLIC SAFETY)

- Consumers verify any product's BIS mark instantly — no more fake certifications
- Grievance portal with auto docket IDs — streamlined complaint tracking
- Hindi + voice input — accessible to 90% non-English speaking citizens
- Reduces accidents from uncertified helmets, cookers, and electricals



ECONOMIC IMPACT (MSME GROWTH)

- MSMEs find correct IS code in seconds — saves weeks of manual research
- Step-by-step certification pathway — reduces compliance costs by 40%+
- Faster certification → faster market entry → increased revenue
- Avoids ₹5 Lakh penalties — protecting small business owners



GOVT IMPACT (DIGITAL ENFORCEMENT)

- 60% less office footfall — citizens self-serve basic BIS queries
- Auto-generated docket IDs — streamlined complaint management
- Data-driven enforcement — track compliance trends by region
- Strengthens "Make in India" quality compliance culture nationwide



TECH IMPACT (DIGITAL INDIA READY)

- GIGW 3.0 compliant — ready for immediate govt website deployment
- RAG architecture — grounded AI responses, not hallucinated
- WhatsApp bot potential — reach 500M+ users on preferred platform
- Open source stack — no vendor lock-in, fully customizable

SOCIAL BENEFITS

- ✓ Consumer awareness about product safety
- ✓ Reduced fake certification in market
- ✓ Empowered citizens to report violations

ECONOMIC BENEFITS

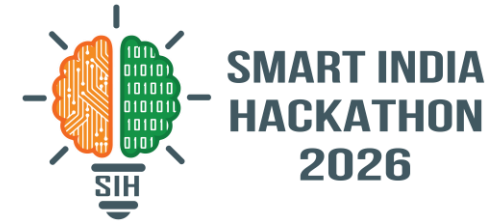
- ✓ MSMEs save compliance costs and time
- ✓ Avoid ₹5 Lakh penalties and legal issues
- ✓ Faster certification → faster market entry

COMPLIANCE BENEFITS

- ✓ Stronger national compliance culture
- ✓ Digital-first reduces paper and office visits
- ✓ Real-time tracking improves enforcement



RESEARCH AND REFERENCES



GOVERNMENT & LEGAL SOURCES

1 Bureau of Indian Standards

<https://bis.gov.in>
Official Indian Standards database — IS codes, QCOs, certification rules

2 BIS Act, 2016

<https://consumeraffairs.gov.in>
Legal framework — penalties up to ₹5 Lakh + 2 years for non-compliance

3 Ministry of Consumer Affairs

<https://consumeraffairs.gov.in>
Policy guidelines, QCOs, grievance redressal mechanisms

4 Manak Online (ISI Portal)

<https://manakonline.in>
Official portal for Scheme-I (ISI Mark) certification applications

5 CRS Portal

<https://crsbis.in>
Official portal for Scheme-II (CRS) registration for electronics

FRAMEWORKS & TOOLS

10 Next.js 16 (App Router)

<https://nextjs.org>
Full-stack React framework with API routes, SSR, Turbopack

11 React 19

<https://react.dev>
UI library with server components and concurrent rendering

12 TypeScript

<https://typescriptlang.org>
Type-safe JavaScript for production-grade code quality

13 Tailwind CSS 4

<https://tailwindcss.com>
Utility-first CSS framework with design tokens and dark mode

14 Vercel Deployment

<https://vercel.com>
Edge network deployment with auto-scaling, SSL, CDN

AI & DATABASE TECHNOLOGIES

6 Google Gemini 3.6 Flash

<https://ai.google.dev>
Multimodal language model powering RAG generation with grounded responses

7 Google text-embedding-004

<https://ai.google.dev>
768-dimensional vector embeddings for Indian Standards semantic search

8 Supabase PostgreSQL

<https://supabase.com>
Cloud database with real-time subscriptions and row-level security

9 pgvector Extension

<https://supabase.com/docs/guides/database/extensions/pgvector>
Vector similarity search — cosine distance for standards retrieval

COMPLIANCE & STANDARDS

15 GIGW 3.0 Guidelines

<https://guidelines.india.gov.in>
Mandatory guidelines for Indian Government Websites and Apps

16 IS 17802:2021

<https://bis.gov.in>
Indian Standard for ICT Accessibility — W3C WCAG 2.1 Level AA

17 Indian Standards Used

IS 4151 • IS 302 • IS 9873 • IS 16333 • IS 2347
Helmets • Electricals • Toys • CRS Electronics • Cookers — verified from bis.gov.in

18 BIS CARE App

[Google Play Store](#)
Official BIS app for consumers to verify CM/L and CRS numbers

ACKNOWLEDGEMENT



THANK YOU

Smart India Hackathon 2026
Problem Statement SIH26107

TEAM STROPHOID

Bennett University | Ministry of Consumer Affairs

*"Making Indian Standards accessible
for every Indian"*

<https://bis-saathi-nine.vercel.app/>